

TARUN L. SANKAR

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EDUCATION

The University of Texas at Dallas - Dallas, TX | Expected May 2027

B.S. in Computer Engineering | Minor in Business Administration

GPA: 3.7/4.0 | Collegium V Honors | Dean's List | National Merit Scholar | Eagle Scout

EXPERIENCE

NorthMark Compute & Cloud - Dallas, TX | Jun 2026 - Aug 2026

Software Engineering Intern, Network Automation

- Built production Python/OpenObserve monitoring for 10+ network engineers, replacing reliance on manual log checks; surfaced 200+ zero-touch provisioning (ZTP) errors in five weeks and flagged Infracore memory pressure, prompting engineers to increase VM RAM.
- Designed a command-driven Infracore snapshot-comparison harness that flags changes from baseline state and posts configuration diffs to Slack.
- Developed a tool-calling diagnostic agent used in network investigations, integrating Jira, Slack, Infracore, and ZTP systems; deployed the service through Harness and Argo-based CI/CD.

University of Texas at Dallas - Dallas, TX | Aug 2024 - Present

Undergraduate Researcher

- Developed a multithreaded POSIX C pipeline on Raspberry Pi to compute S4 scintillation indices every five minutes from GNSS data sampled at up to 20 Hz; the system has operated unattended for over a year.
- Validated indices from the low-cost system against a reference instrument over three days, achieving $R^2 = 1.00$ before downsampling; integrated Azure Blob Storage and Dash for storage and visualization.

MyIntent.io - Remote | Aug 2024 - Nov 2024

Software Engineering Intern, Full Stack Development

- Built the document-upload UI for a Next.js/NestJS RFP automation tool that generates business documents from structured customer inputs.
- Implemented document storage and retrieval for retrieval-augmented generation (RAG), enabling business-document generation using uploaded source material.

PROJECTS

Orune - Generative Educational Animation | Aug 2026

useorune.com

- Built and launched a React/TypeScript, Express and Python app used by TikTok creators to generate narrated Manim videos in 1-3 minutes at approximately \$0.60 operating cost per video.
- Implemented automatic render retries, frame annotations and model-assisted review to support video revisions while preserving immutable render history.
- Queued renders through Cloud Tasks for isolated E2B workers, stored artifacts privately in Google Cloud Storage, and implemented usage credits with Stripe billing.

Cinder Control - TAMUHack 2026 | Jan 2026

2nd Place, NorthMark HPC Sponsor Track; Best Solo Project

- Built a wildfire simulation prototype solo in 36 hours using Next.js and Python/FastAPI.
- Modeled fire spread with NumPy using fuel, wind and slope; visualized scenarios in Mapbox and flagged nearby infrastructure using OpenStreetMap data.
- Integrated OpenAI and ElevenLabs for voice/text scenario exploration and NOAA/NWS weather data for the simulation interface.

RESEARCH PRESENTATION

First Author, Poster - "Real-Time Scintillation Monitoring using Low-Cost GNSS-Based Monitors."

2025 Joint CEDAR/GEM Workshop, Des Moines, IA.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL, Bash

Web & Backend: React, Next.js, Node.js, Express, NestJS, FastAPI, PostgreSQL

Cloud & DevOps: Google Cloud, Azure, Docker, Kubernetes, Terraform, Harness, Argo, OpenObserve

AI & LLMs: OpenAI API, RAG, LLM tool calling, Model Context Protocol (MCP) tool development

AI-Assisted Development: Claude, Codex, GitHub Copilot

Scientific & Media: NumPy, Dash, Manim, FFmpeg